

Security+ 4.5 · Monitoring Data

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.5** —
Monitoring Data (7:25) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Data monitoring tools prevent data loss and detect tampering. DLP prevents sensitive data from leaving via email/USB/upload; FIM alerts on hash changes in critical files; USB blocking disables external storage; DLP in use (confidential computing).

High-Yield Core Concepts Tested on Exam

- DLP: monitors and blocks data leaving via email/USB/upload
- FIM: alerts on any modification to critical files
- USB blocking: prevents data exfiltration via removable media
- DLP can be applied to data in use (confidential computing)
- FIM uses file hashes to detect changes

Exam Tips & Classic Scenarios

- 'prevents sensitive data from leaving the organization' = DLP

Typical Exam Scenario: A company blocks sending a spreadsheet containing Social Security numbers via email. This is an example of DLP.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.5 — Monitoring Data)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **DLP:** Data Loss Prevention: Software/hardware safeguards monitoring, identifying, and preventing unauthorized exfiltration of sensitive organizational assets in use, transit, or rest.

 Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. A control that prevents an employee from sending a confidential spreadsheet outside the company is:

- A)** DLP (Data Loss Prevention)
- B)** SSO
- C)** MFA
- D)** VPN

2. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'DLP: monitors and BLOCKS data leaving via email/USB/upload.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: DLP: monitors and BLOCKS data leaving via email/USB/upload...
- D)** Allow shared access among administrators using generic credentials

3. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'FIM: alerts on any modified critical file'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: FIM: alerts on any modified critical file...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'prevents sensitive data from leaving the organization = DLP.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'prevents sensitive data from leaving the organization' = DLP...
- D)** Allow shared access among administrators using generic credentials

5. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'company blocks sending spreadsheet with SSN via email'=DLP. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'company blocks sending spreadsheet with SSN via email'=DLP
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

6. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'DLP: monitors and BLOCKS data leaving via email/USB/upload'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: DLP: monitors and BLOCKS data leaving via email/USB/upload...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

7. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'FIM: alerts on any modified critical file'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: FIM: alerts on any modified critical file...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: "prevents sensitive data from leaving the organization'=DLP'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'prevents sensitive data from leaving the organization'=DLP...

9. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'company blocks sending spreadsheet with SSN via email'=DLP. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'company blocks sending spreadsheet with SSN via email'=DLP...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

10. Corporate scenario (Monitoring Data): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'DLP: monitors and BLOCKS data leaving via email/USB/upload'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: DLP: monitors and BLOCKS data leaving via email/USB/upload...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — DLP (Data Loss Prevention).

✓ **Why it's correct:** DLP (Data Loss Prevention) is a set of technologies and policies aimed at preventing the exfiltration of confidential data by monitoring and controlling transfers of sensitive information. In CompTIA Security+ (SY0-701), DLP is presented as a control that can block the sending of data outside the organization, whether via email, web, USB, or other channels. The question describes exactly a control that prevents an employee from sending a confidential spreadsheet outside the company, which is the primary function of DLP.

✗ Why the others are wrong:

- **SSO:** SSO (Single Sign-On) is an authentication mechanism that allows a user to access multiple systems with a single credential. It does not control data transfer or prevent information exfiltration. Therefore, it does not apply to the described scenario.
- **MFA:** MFA (Multi-Factor Authentication) is an authentication method that requires two or more verification factors to grant access. It strengthens authentication but does not prevent an authorized user from sending confidential data outside. It is not a data loss prevention control. Thus, it is not the correct answer.
- **VPN:** VPN (Virtual Private Network) is a technology that creates an encrypted tunnel for secure communication, but it does not have the function of inspecting or blocking the transfer of confidential data. It protects transmission but does not prevent exfiltration by an authorized user. Therefore, it does not meet the requirement of the question.

2) C — Implement the corresponding solution: DLP: monitors and BLOCKS data leaving via email/USB/upload....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): DLP: monitors and BLOCKS data leaving via email/USB/upload. Data Loss Prevention (DLP) solutions monitor and control data in motion, at rest, and in use, preventing unauthorized exfiltration of sensitive information through channels such as email, USB devices, or cloud uploads.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

3) B — Implement the corresponding solution: FIM: alerts on any modified critical file....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): FIM: alerts on any modified critical file. File Integrity Monitoring (FIM) is a control that monitors and detects changes to critical system files, configuration files, and content files. It works by creating a baseline of file attributes (such as cryptographic hashes) and then periodically comparing the current state against the baseline. Any modification triggers an alert, enabling security teams to detect unauthorized changes that could indicate malware, insider threats, or configuration drift. FIM is essential for compliance (e.g., PCI DSS, HIPAA) and is a key

detective control.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

4) C — Implement the corresponding solution: 'prevents sensitive data from leaving the organization' = DLP....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): DLP (Data Loss Prevention) is a set of technologies and policies that detect and prevent unauthorized exfiltration of sensitive data. It can monitor data in use, in motion, and at rest.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) B — Implement the corresponding solution: 'company blocks sending spreadsheet with SSN via email' = DLP.

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): 'company blocks sending spreadsheet with SSN via email' = DLP. Data Loss Prevention (DLP) inspects and blocks sensitive data exfiltration.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: DLP: monitors and BLOCKS data leaving via email/USB/upload....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for

lesson 4.5 (Monitoring Data): DLP: monitors and BLOCKS data leaving via email/USB/upload.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) A — Implement the corresponding solution: FIM: alerts on any modified critical file....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): FIM: alerts on any modified critical file. File Integrity Monitoring (FIM) is a detective control that alerts on changes to critical files, helping to detect unauthorized modifications.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) D — Implement the corresponding solution: 'prevents sensitive data from leaving the organization'=DLP...

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): 'prevents sensitive data from leaving the organization'=DLP.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) A — Implement the corresponding solution: 'company blocks sending spreadsheet with SSN via email'=DLP...

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): 'company blocks sending spreadsheet with SSN via email'=DLP.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) B — Implement the corresponding solution: DLP: monitors and BLOCKS data leaving via email/USB/upload....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Monitoring Data): DLP: monitors and BLOCKS data leaving via email/USB/upload. DLP (Data Loss Prevention) solutions inspect and control data flows to prevent unauthorized exfiltration of sensitive information.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.